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Division : CS-1

Batch : C13

Course : EDS

Essential of Data Science Laboratory

Practical 1:

1.1.1 Calculate Momentum:

The screenshot displays the CODETANTRA web-based IDE. The interface is divided into several sections:

- Header:** CODETANTRA logo, navigation links (Home, Learn Anywhere), user information (202501040051@mitaoe.ac.in), and a Logout button.
- Left Panel (Task Description):**
 - 1.1.1. Calculate Momentum:** A task card with a timer (07:19) and icons for a file, moon, and share.
 - Description:** "Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula: $p = m \times v$ where: m is the mass of the object (in kilograms). v is the velocity of the object (in meters per second)."
 - Input Format:**
 - A single floating-point number representing the mass of the object in kilograms.
 - A single floating-point number representing the velocity of the object in meters per second.
 - Output Format:**
 - The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).
 - Sample Test Cases:** A button to expand the test cases section.
- Right Panel (Code Editor and Test Results):**
 - Explorer:** A file tree showing a file named "calculate...".
 - Code Editor:** Contains the following Python code:

```
1 mass = float(input())
2 velocity = float(input())
3 momentum = mass*velocity
4 print(f"momentum: {momentum:.2f}kgm/s")
```
 - Test Results:**
 - Average time:** 0.014 s (13.75 ms)
 - Maximum time:** 0.019 s (19.00 ms)
 - Status:** 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed.
 - Test Case 1:** Expected output 5.0, 10.0, 50.00kgm/s; Actual output 5.0, 10.0, 50.00kgm/s. Time: 11 ms.
 - Test Case 2:** Time: 10 ms.
 - Terminal:** A button to view the terminal output.
 - Test cases:** A button to manage test cases.
- Footer:** Navigation buttons: < Prev, Reset, Submit, Next >.

1.1.2 Conditional Calculations:

1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:
• The input consists of a single integer n .

Output Format:
• If n is a single-digit number, print its square.
• If n is a two-digit number, print its square root (rounded to two decimal places).
• If n is a three-digit number, print its cube root (rounded to two decimal places).
• Else print "Invalid".

Sample Test Cases

```
1 n = int(input())
2
3 if 1 <= n <= 9:
4     print(n*n)
5 elif 10 <= n <= 99:
6     print(f"{n**(1/2) :.2f}")
7 elif 100 <= n <= 999:
8     print(f"{n**(1/3) :.2f}")
9 else:
10    print("Invalid")
```

Average time: 0.006 s (5.86 ms) | Maximum time: 0.010 s (10.00 ms)

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case	Expected output	Actual output
Test case 1	81	81
Test case 2		
Test case 3		

1.1.3 Days Between two Dates:

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:
• The first line contains the first date in the format $YYYY-MM-DD$.
• The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:
• An integer representing the number of days between the given dates.

Note:
• The first date should always be considered earlier than the second date.

Sample Test Cases

```
1 from datetime import date
2 date1 = input().strip()
3 date2 = input().strip()
4 d1 = date.fromisoformat(date1)
5 d2 = date.fromisoformat(date2)
6
7 diff = (d2-d1).days
8 print(diff)
```

Average time: 0.013 s (12.75 ms) | Maximum time: 0.020 s (20.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case	Expected output	Actual output
Test case 1	2014-07-02 2014-11-02 123	2014-07-02 2014-11-02 123
Test case 2		

1.1.4 Reverse a Number:

1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2 rev = 0
3 while n > 0:
4     digit = n % 10
5     rev = rev*10 + digit
6     n = n // 10
7 print(rev)
```

Average time
0.011 s
11.00 ms

Maximum time
0.020 s
20.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 13 ms Debug

Expected output	Actual output
5367	5367
7635	7635

Test case 2 8 ms

Terminal Test cases

< Prev Reset Submit Next >

1.1.5 Multiplication Table:

1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 n = int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n*i}")
4
5
6
7
```

Average time
0.005 s
5.00 ms

Maximum time
0.007 s
7.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal Test cases

< Prev Reset Submit Next >

1.2.1 Pass or Fail:

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1.2.1. Pass or Fail

38.53

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

Submit

Explorer

Debugger

```
1 n = int(input())
2 marks = list(map(int, input().split()))
3 if any (m<40 for m in marks):
4     print("Fail")
5 else:
6     ave = sum(marks)/n
7     print("Aggregate Percentage: %.2f"%ave)
8     if ave>75:
9         print("Grade: Distinction")
10    elif 60 <= ave < 75:
11        print("Grade: First Division")
12    elif 50 <= ave < 60:
13        print("Grade: Second Division")
14    elif 40 <= ave < 50:
15        print("Grade: Third Division")
```

Average time

0.008 s

8.20 ms

Maximum time

0.017 s

17.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

17 ms

Debug

Expected output

4

55 65 78 89

Actual output

4

55 65 78 89

Terminal

Test cases

< Prev

Reset

Submit

Next >

1.2.2 Fibonacci Series:

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1.2.2. Fibonacci Series

16.47

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

Submit

Explorer

Debugger

```
1 def fibonacci(n):
2
3     if n==1:
4         return 0
5     elif n==2:
6         return 1
7     else:
8         return fibonacci(n-1) + fibonacci(n-2)
9
10
11 n = int(input())
12 for i in range(1, n + 1):
13     print(fibonacci(i), end=" ")
14
```

Average time

0.016 s

16.00 ms

Maximum time

0.040 s

40.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

7 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Terminal

Test cases

< Prev

Reset

Submit

Next >

1.2.3 Pattern-1:

1.2.3. Pattern - 1 06:33

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

```
1 n = int(input())
2
3 for i in range(1,n+1):
4     for j in range(1,i+1):
5         print("*",end=" ")
6     print()
```

Average time: 0.009 s, 8.50 ms | Maximum time: 0.014 s, 14.00 ms

2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 14 ms Debug

Expected output	Actual output
5	5
* *	* *
* * *	* * *
* * * *	* * * *
* * * * *	* * * * *

Terminal Test cases

< Prev Reset Submit Next >

1.2.4 Pattern - 2 :

1.2.4. Pattern - 2 08:19

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

```
1 n = int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
```

Average time: 0.006 s, 6.50 ms | Maximum time: 0.010 s, 10.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output	Actual output
5	5
1 *	1 *
1 2 *	1 2 *
1 2 3 *	1 2 3 *
1 2 3 4 *	1 2 3 4 *
1 2 3 4 5 *	1 2 3 4 5 *

Terminal Test cases

< Prev Reset Submit Next >

